MARCAINE 0.50 PER CENT (100 mg/20 ml) ADRENALINE, injectable solution.

QUALITATIVE AND QUANTITATIVE COMPOSITION

Anhydrous bupivacaine hydrochloride		100.00 mg
(in the form of bupivacaine hydrochloride)		
Adrenaline		0.10 mg
Sodium chloride		160.00 mg
Sodium metabisulfite		10.00 mg
Hydrochloric acid/sodium hydroxide	qs.pH =	3.3 to 5.0
Water for injectable preparations		qsp 20 ml

PHARMACEUTICAL FORM

Bupivacaine is classed as a membrane stabilising agent and is a local anaesthetic of the amide type. Marcain solutions are available with or without adrenaline, contain no antimicrobial agent and should be used only once and any residue discarded.

Marcain with adrenaline solutions for injection are sterile, isotonic aqueous solutions of bupivacaine hydrochloride in Water for Injections BP. The pH of the solution is adjusted with sodium hydroxide or hydrochloric acid to remain between 3.3 - 5.0. They contain sodium metabisulfite as an antioxidant.

CLINICAL PARTICULARS

THERAPEUTIC INDICATIONS

Marcain solutions are indicated for the production of local or regional anaesthesia and analgesia in individuals as follows:

Surgical anaesthesia

- Epidural block for surgery
- Field block (minor and major nerve blocks and infiltration).

Analgesia

- Continuous epidural infusion or intermittent bolus epidural administration for analgesia in postoperative pain or labour pain.
- Field block (minor nerve block and infiltration).

POSOLOGY AND METHOD OF ADMINISTRATION

Marcain with Adrenaline should only be used by clinicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used.

In order to prevent inadvertent intravascular injections, it is important that great caution be exercised. Careful aspiration before and during the injection of the total dose is recommended. The total dose must be injected slowly, 25-50 mg/min or in divided doses with continuous verbal contact being maintained with the patient and controls of heart rate. In epidural injection of high doses, a test dose of 3-5 ml Marcain adrenalin is recommended. An inadvertent intravascular injection can cause e.g. a brief increase in heart rate and inadvertent intrathecal injection can produce signs of spinal block. If toxic symptoms occur, the injection must be discontinued immediately.

Bupivacaine should only be used by or under the supervision of doctors experienced in local or regional anaesthesia techniques.

Bupivacaine hydrochloride is available in adrenaline or non-adrenaline forms and in different presentations.

The form, concentration and presentation used vary as a function of the indication and the ultimate objective (surgical anaesthesia or pure analgesia) and the age and pathological state of the patient.

The use of adrenaline forms prolongs the period of action. The more concentrated forms produce a more constant and more intense motor block.

The following are average posologies given for information:

Surgical anaesthesia:
Peridural:
Adults: 0.50 per cent (5 mg/ml) bupivacaine hydrochloride: 6 to 8 mg per segment, not exceeding 12 to 24 ml in total.
Caudal:
Children: 0.25 per cent (2.5 mg/ml) bupivacaine hydrochloride: 0.6 – 0.8 ml/kg = 1.5 – 2 mg/kg
Adults: 0.50 per cent (5 mg/ml) bupivacaine hydrochloride: 15 to 30 ml.
Plexic blocks:
0.50 per cent (5 mg/ml) bupivacaine hydrochloride preferably with adrenaline: 20 to 30 ml.
0.25 per cent (2.5 mg/ml) bupivacaine hydrochloride preferably with adrenaline: 25 to 40 ml.
Truncular blocks:
0.25 per cent (2.5 mg/ml) bupivacaine hydrochloride or 0.5 per cent (5 mg/ml) bupivacaine hydrochloride: a few ml to 15 or 20 ml depending on the nerve.
Intercostal block:
0.25 per cent (2.5 mg/ml) bupivacaine hydrochloride: 1 to 2 ml per nerve (never exceed the 100 mg dose, i.e. 40 ml).

Obstetric anaesthesia:
The doses are initial doses, which if required may be repeated every two to three hours.
Epidural anaesthesia (for vaginal delivery and vacuum extraction): 6-8-10 ml (15-50 mg bupivacaine hydrochloride).

Caudal anaesthesia (for vaginal delivery and vacuum extraction): 6-8-10 ml (15-50 mg bupivacaine hydrochloride).
Pudendal block (on each side): 5-10 ml (12.5-50 mg bupivacaine hydrochloride).

Treatment of pain: 0.25 per cent (2.5 mg/ml) bupivacaine hydrochloride.
Peridural analgesia: 5 to 15 ml to be renewed approximately every 6 hours.
Various blocks: 8 to 20 ml.

Maximum recommended doses
Marcain with Adrenaline 2.5 mg/ml + 5 micrograms/ml: 60 ml (150 mg bupivacaine hydrochloride)

Marcain with Adrenaline 5 mg/ml + 5 micrograms/ml: 30 ml (150 mg bupivacaine hydrochloride).

The maximum recommended dose on one and the same occasion is calculated on the basis of 2 mg/kg bodyweight, and for adults is a maximum of 150 mg within a four-hour period. The maximum permissible dose during a 24-hour period is 400 mg (80 ml Marcain with Adrenaline 5 mg/ml + 5 micrograms/ml). The total dose must be corrected on the basis of the patient's age, constitution and other relevant circumstances.

Do not reuse a bottle once opened.

CONTRAINDICATIONS
Hypersensitivity to the active substance, local anaesthetics of the amide type or to any of the excipients. Bupivacaine must not be used for intravenous regional anaesthesia (Bier's block). Bupivacaine must not be used for epidural anaesthesia in patients with pronounced hypotension such as in cardiogenic or hypovolaemic shock.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE
Regional or local anaesthetic procedures, except those of the most trivial nature, should always be carried out in the proximity of resuscitation equipment. Before major blocks, an intravenous cannula should be inserted before the local anaesthetic is injected.

There have been reports of cardiac arrest or deaths in cases of use of bupivacaine for epidural anaesthesia or peripheral nerve block. In some cases, resuscitation has been difficult or impossible despite adequate treatment. Extensive peripheral nerve blocks can involve large volumes of local anaesthetics being administered to richly vascularised areas, often in the proximity of large blood vessels. In these cases, there is an increased risk of intravascular injection and/or systemic absorption, which can lead to high plasma concentrations.

Like all local anaesthetic agents, bupivacaine can cause acute central nervous and cardiovascular toxic effects in cases of use leading to high concentrations in the blood. This applies particularly after inadvertent intravascular administration or injection into highly vascularised areas.

Some regional anaesthesia techniques may be associated with severe adverse reactions, as indicated below:

- Epidural anaesthesia can cause cardiovascular depression, especially in cases of concomitant hypovolaemia. Caution should therefore be exercised in patients with impaired cardiovascular function.
- Retrobulbar injections can in rare cases reach the cranial subarachnoid space and cause e.g. temporary blindness, cardiovascular collapse, apnoea and convulsions. These symptoms must be treated immediately.
- Retro- and peribulbar injections with local anaesthetics may involve some risk of persistent ocular muscle dysfunction. The prime causes are traumatic nerve injury and/or local toxic effects on muscles and nerves by injected local anaesthetic. The extent of this tissue damage depends on the degree of trauma, the concentration of the local anaesthetic, and the duration of exposure to the local anaesthetic. For this reason, the lowest effective dose should be chosen. Vasoconstrictors may aggravate tissue reactions and should be used only when indicated. Inadvertent intravascular injection in the head and neck regions can cause immediate cerebral symptoms even with low doses.
- Paracervical block can sometimes cause bradycardia/tachycardia in the foetus, and the foetus's heart rhythm must be closely monitored.

Caution should be exercised in patients with degree II or III AV block since local anaesthetics can lower the conduction capacity of the myocardium. Elderly patients and patients with severe hepatic disease, severely impaired renal function or with generally reduced general condition also require special attention.

Patients treated with class III anti-arrhythmic drugs (e.g. amiodarone) should be closely observed and ECG monitoring should be considered, since the cardiac effects of bupivacaine and class III anti-arrhythmic drugs can be additive.

Epidural anaesthesia can lead to hypotension and bradycardia. The risk of such effects can be reduced, e.g. by injecting a vasopressor. Hypotension should be treated immediately with a sympathomimetic intravenously, repeated as necessary.

Adrenaline-containing solutions should be used with caution in patients with severe or untreated hypertension, poorly controlled thyrotoxicosis, ischaemic heart disease, AV block, cerebrovascular insufficiency, advanced diabetes, and other conditions that can be exacerbated by adrenaline. Caution should also be exercised in cases of use in peripheral parts of the body such as fingers, or parts of the body that for other reasons have a low blood supply. In cases of administration of high doses the risk of systemic effects of adrenaline should be taken into account.

Sodium metabisulfite (anti-oxidant) can in rare cases cause allergic reactions, e.g. asthma of varying severity and anaphylactic reactions.

There have been post-marketing reports of chondrolysis in patients receiving post- operative intra-articular continuous infusion of local anaesthetics. The majority of reported cases of chondrolysis have involved the shoulder joint. Due to multiple contributing factors and inconsistency in the scientific literature regarding mechanism of action, causality has not been established. Intra-articular continuous infusion is not an approved indication for Marcaine Adrenaline.

Hepatic dysfunction, with reversible increases of alanine aminotransferase (ALT), alkaline phosphates (AlkP) and bilirubin, has been observed following repeated injections or long-term infusions of bupivacaine. Association between bupivacaine use and the development of drug-induced liver injury (DILI) has been reported in a small number of literature reports especially with prolonged use. While the pathophysiology of this reaction remains unclear, immediate withdrawal of bupivacaine has shown rapid clinical improvement. If signs of hepatic dysfunction are observed during administration with bupivacaine, the medicinal product should be discontinued. (See section Undesirable effects).

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Bupivacaine should be used with caution with other local anaesthetics or drugs that are structurally similar to local anaesthetics, i.e. class IB anti-arrhythmic drugs, as the systemic toxic effects are additive.

No specific interaction studies with local anaesthetics and class III anti-arrhythmic drugs (e.g. amiodarone) have been carried out, but caution is recommended (See also Special warnings and precautions for use).

Non-selective beta-blockers

Non-selective beta-blockers such as propranolol enhance the pressor effect of *adrenaline*, which may lead to pronounced hypertension and bradycardia. The combination might need dose adjustment.

Anaesthetics for inhalation

Adrenaline can cause severe cardiac arrhythmias when injected during general anaesthesia with *halothane*. The combination might need dose adjustment.

Tricyclic antidepressants

The pressor effect of *adrenaline* can, in combination with tricyclic antidepressants, cause a marked increase in blood pressure. In short-term studies with high-dose intravenous adrenaline the effect has been shown to be increased 2- to 3-fold.

A combination of *adrenaline-containing* solutions and *oxytocic* drugs of the ergot type can cause a marked, persistent increase in blood pressure and may possibly cause cerebrovascular and cardiac accidents.

Phenothiazine derivatives and *butyrophenone derivatives* can decrease or inhibit the pressor effect of *adrenaline.*

FERTILITY, PREGNANCY AND LACTATION

PREGNANCY

In cases of paracervical block the risk of reactions such as bradycardia/ tachycardia in the foetus increases, therefore foetal heart rate must be closely monitored.

It should also be noted that the addition of adrenaline can reduce uterine perfusion and uterine contractions during parturition, especially in cases of intravenous injection. See also Pharmacokinetic properties.

LACTATION

Bupivacaine passes into breast milk, but the risk of this affecting the child appears unlikely with therapeutic doses. It is not known whether *adrenaline* passes into breast milk, but the risk of an effect on the child appears unlikely with therapeutic doses.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Depending on the dose and method of administration, bupivacaine can have a transient effect on movement and coordination.

UNDESIRABLE EFFECTS

Undesirable effects caused by the product itself can be difficult to distinguish from the physiological effects of the nerve block (e.g. fall in blood pressure, bradycardia), events caused directly by the needle puncture (e.g. nerve injury) or caused indirectly by the needle puncture (e.g. epidural abscess).

Neurological damage is a rare but well recognised consequence of regional, and particularly epidural and spinal anaesthesia.

Organ System	**Frequency**	**Symptom**
Immune system disorders	Rare (≥1/10 000, < 1/1000)	Allergic reactions, anaphylactic reaction/shock
Nervous system disorders	Common (≥1/100, < 1/10)	Paraesthesia, dizziness
	Uncommon (≥1/1000, < 1/100)	Signs and symptoms of CNS toxicity (convulsions, paraesthesia circumoral, numbness of the tongue, hyperacusis, visual disturbances, loss of consciousness, tremor, light headedness, tinnitus, dysarthria)
	Rare (≥1/10 000, < 1/1000)	Neuropathy, peripheral nerve injury, arachnoiditis, paresis and paraplegia
Eye disorders	Rare (≥1/10 000, < 1/1000)	Diplopia
Cardiac	Common (≥1/100, < 1/10)	Bradycardia

disorders		
	Rare (≥1/10 000, < 1/1000)	Cardiac arrest, cardiac arrhythmias
Vascular disorders	Very common (≥1/10)	Hypotension
	Common (≥1/100, < 1/10)	Hypertension
Respiratory, thoracic and mediastinal disorders	Rare (≥1/10 000, < 1/1000)	Respiratory depression
Gastrointestinal disorders	Very common (≥1/10)	Nausea
	Common (≥1/100, < 1/10)	Vomiting
Renal and urinary disorders	Common (≥1/100, < 1/10)	Urinary retention
Hepatobiliary disorders	Not known (cannot be estimated from the available data)	Impaired hepatic function/increase in ALT*

**Hepatic impairment, with reversible increases in ALT, AlkP and bilirubin, have been observed following repeated bupivacaine injections or bupivacaine infusions of long duration. If signs of hepatic impairment are observed during treatment with bupivacaine, the medicinal product must be discontinued (See section Special warnings and precautions for use).*

<u>OVERDOSE AND TREATMENT</u>

SYMPTOMS AND SIGNS

Acute systemic toxicity:

Systemic toxic reactions include the central nervous system and the cardiovascular system. Such reactions are caused by a high concentration of local anaesthetic in the blood, which can occur as a result of accidental intravascular injection, overdose, or unusually rapid absorption from richly vascularised tissues (see also Special warnings and precautions for use).

CNS symptoms are similar for all local anaesthetics of the amide type, while cardiac symptoms differ for different drugs, both quantitatively and qualitatively.

Accidental intravascular injections of local anaesthetics can cause immediate system-toxic reactions (within seconds to a few minutes). Signs of systemic toxicity in cases of overdose occur later (15-60 minutes after injection) due to a slower increase in the concentration of local anaesthetic in the blood.

CNS toxicity occurs gradually, with symptoms and reactions of escalating severity. The first symptoms are usually light-headedness, circumoral paraesthesia, numbness of the tongue, hyperacusis, tinnitus and visual disturbances. Difficulty in articulating, muscle twitching or tremor are more serious and precede generalised convulsions. These signs must not be interpreted as neurotic behaviour. Unconsciousness and grand mal seizures may follow and may last from a few seconds to several minutes. Hypoxia and hypercapnia occur rapidly during the seizures due to the increased muscular activity and inadequate ventilation. In severe cases apnoea may occur. Acidosis increases the toxic effects of local anaesthetics.

Recovery is due to the metabolism and distribution of the local anaesthetic away from the central nervous system. This takes place rapidly unless very large amounts of the drug have been injected.

Cardiovascular effects generally cause a more serious situation and are usually preceded by signs of CNS toxicity, which can, however, be masked by general anaesthesia or heavy sedation with drugs such as benzodiazepines or barbiturates. A fall in blood pressure, bradycardia, arrhythmia and even cardiac arrest can occur as a result of high systemic concentrations of local anaesthetics. Cardiovascular toxic effects are often related to depression of the conduction system in the heart and myocardium, leading to reduced cardiac output, hypotension, AV block, bradycardia and sometimes ventricular arrhythmias including ventricular tachycardia, ventricular fibrillation and cardiac arrest. These states are often preceded by signs of severe CNS toxicity, e.g. in the form of convulsions, but in rare cases cardiac arrest has occurred without prodromal CNS effects. After a very rapid intravenous bolus injection, such a high blood concentration of bupivacaine can be reached in the coronary vessels that an effect on the circulation occurs alone or before effects on the CNS. With this mechanism, myocardial depression can then occur even as a first symptom of intoxication.

As children often develop greater blocks only after anaesthesia is begun, particular alertness to early signs of intoxication is required in this group.

Systemic effects of adrenaline must also be taken into account in cases of overdose.

TREATMENT

In the event of total spinal block optimal ventilation should be secured (free airways, oxygen, if necessary intubation and assisted ventilation). In the event of hypotension/bradycardia a vasopressor with inotropic effect should be administered.

If signs of acute systemic toxicity occur the administration of local anaesthetics must be stopped immediately and CNS symptoms (convulsion, CNS depression) must promptly be treated with appropriate airway / respiratory support and the administration of anticonvulsant drugs.

If circulatory shock occurs, (hypotension, bradycardia), appropriate treatment with intravenous fluids, vasopressors, inotropic substances and/or other lipid emulsions should be considered. Children must be given doses in proportion to their age and bodyweight for treatment of systemic toxicity.

In the event of circulatory arrest, immediate cardiopulmonary resuscitation should be instituted. Optimal oxygenation and ventilation and circulatory support as well as treatment of acidosis are of vital importance.

Should cardiac arrest occur, a successful outcome may require prolonged resuscitative efforts.

PHARMACOLOGICAL PROPERTIES

PHARMACODYNAMICS

Pharmacotherapeutic class:
Local anaesthetic

ATC code: N01B B51

Marcain with Adrenaline contains bupivacaine, which is a long-acting local anaesthetic of the amide type, and the vasoconstrictor adrenaline (epinephrine). Bupivacaine reversibly blocks impulse conduction in the nerves by inhibiting the transport of sodium ions through the nerve membrane. Similar effects can also be seen on excitatory membranes in the brain and myocardium.

The most prominent property of bupivacaine is its long duration of effect, and the difference in duration between bupivacaine with and without adrenaline is relatively small. Lower concentrations produce lesser effects on motor nerve fibres and a shorter duration of effect, and can be suitable for prolonged pain relief, e.g. during parturition or postoperatively.

PHARMACOKINETICS

The rate of absorption is dependent on dose, route of administration and perfusion at the injection site. The addition of adrenaline can reduce the rate of absorption, whereas the time to peak plasma concentration is little affected. The effect varies with the type of block, dose and concentration. In adults, adrenaline decreases peak plasma concentrations by up to 50% in brachial plexus block and by 5 – 25% in epidural block. Intercostal blockades produce the highest plasma concentrations (1-4 mg/l) after a dose of 400 mg), due to rapid absorption, while subcutaneous injections in the abdomen produce the lowest plasma concentrations. In children rapid absorption and high plasma concentrations are seen in cases of caudal block (approx. 1-1.5 mg/l after a dose of 3 mg/kg).

Bupivacaine is absorbed completely and bi-phasically from the epidural space with half-lives of approx. 7 minutes and 6 hours respectively. The slow absorption is rate-limiting in the elimination of bupivacaine, which explains why the elimination half-life after epidural administration is longer than after intravenous administration.

The volume of distribution of bupivacaine in steady state is 73 litres, the degree of hepatic extraction is 0.40, total plasma clearance is 0.58 l/min and the elimination half-life is 2.7 hours.

The elimination half-life in newborns is up to 8 hours longer than in an adult. In children over 3 months the half-life is equivalent to that in adults.

Plasma protein binding is 96 % and binding is predominantly to alpha-1- glycoprotein. After a major operation the level of this protein may be increased and give a higher total plasma concentration of bupivacaine. However, the unbound concentration of bupivacaine remains the same. This explains why plasma concentrations above toxic levels can be well tolerated.

Bupivacaine is metabolised almost completely in the liver, predominantly through aromatic hydroxylation to 4-hydroxybupivacaine and N-dealkylation to pipecholylxylidine, which are both mediated by cytochrome P450 3A4. Clearance is thus dependent on hepatic perfusion and the activity of the metabolising enzyme.

Bupivacaine crosses the placenta and the concentration of free bupivacaine is the same in the mother and the foetus. However, the total plasma concentration is lower in the foetus, which has a lower degree of protein binding.

PHARMACEUTICAL PARTICULARS

INCOMPATIBILITIES

Alkalisation can cause precipitation, as bupivacaine is limitedly soluble at pH above 6.5.
In addition, alkaline solutions can cause rapid degradation of adrenaline.

SHELF-LIFE

The expiry date is indicated on the packaging.

STORAGE CONDITIONS

Store below 25°C.
However, the product may be kept during transport for a period not exceeding 5 days without special storage precautions.

NATURE AND CONTENTS OF CONTAINER

20 ml glass vial

Pack Size

5’s x 20mL

INSTRUCTIONS FOR USE / HANDLING

Due to instability of adrenaline, the product should not be sterilised.

The solution for injection must not be kept in such a way that it can act on metals, e.g. cannulas or syringes with metal parts. Metal ions may be released, which can cause swelling in the area of the injection and more rapid degradation of the adrenaline.

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